

4th June 2026

WMH SEGMENTATION

Computational Intelligence Lab

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AGENDA

01. Problem definition

02. Data source

03. Conceptual Framework

04. Feature Computation

05. Feature significance
analysis

06. Fetaure visualization

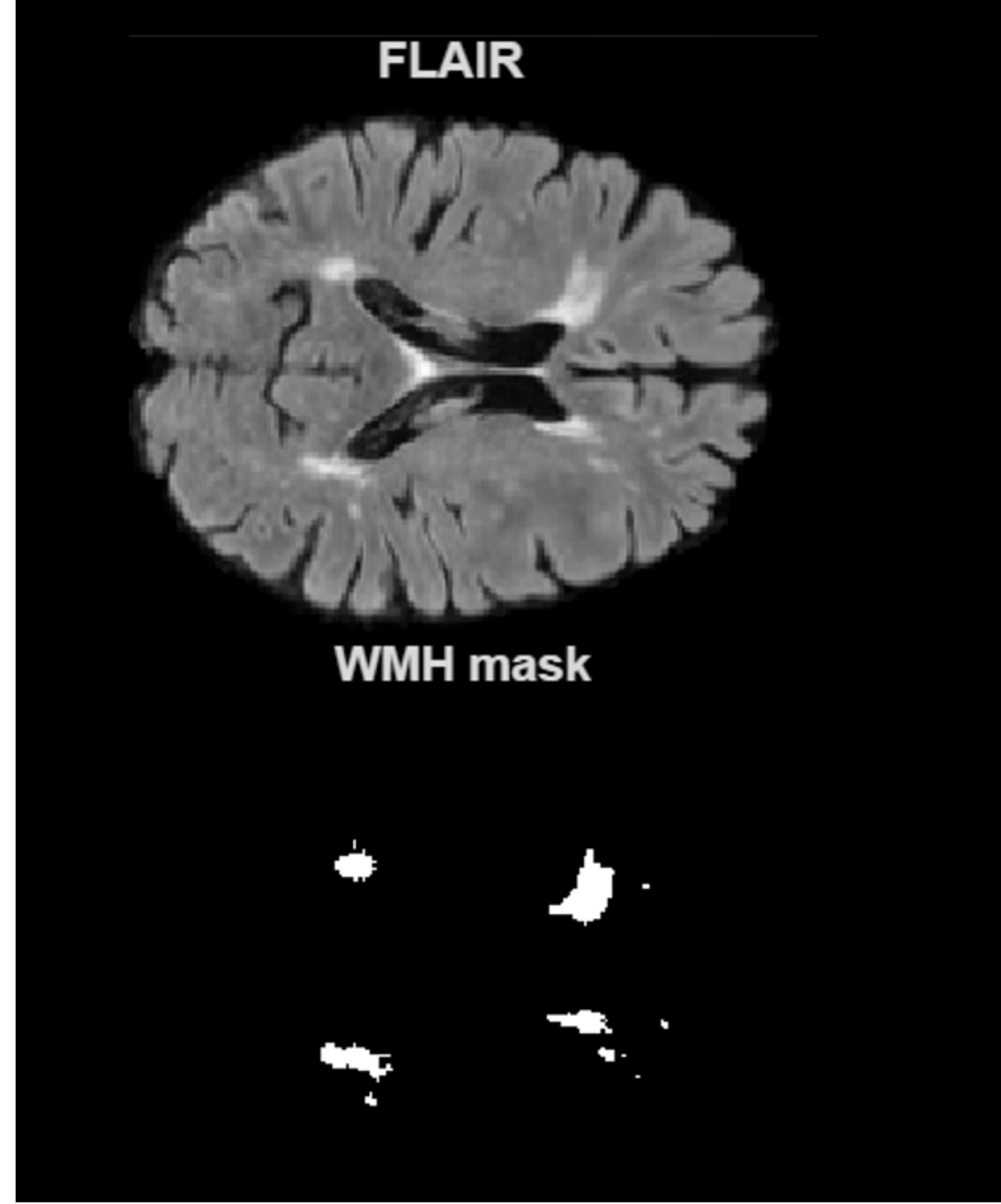
07. Preprocessing

08. Mamdani FIS

09. Genetic Optimization

PROBLEM DEFINITION

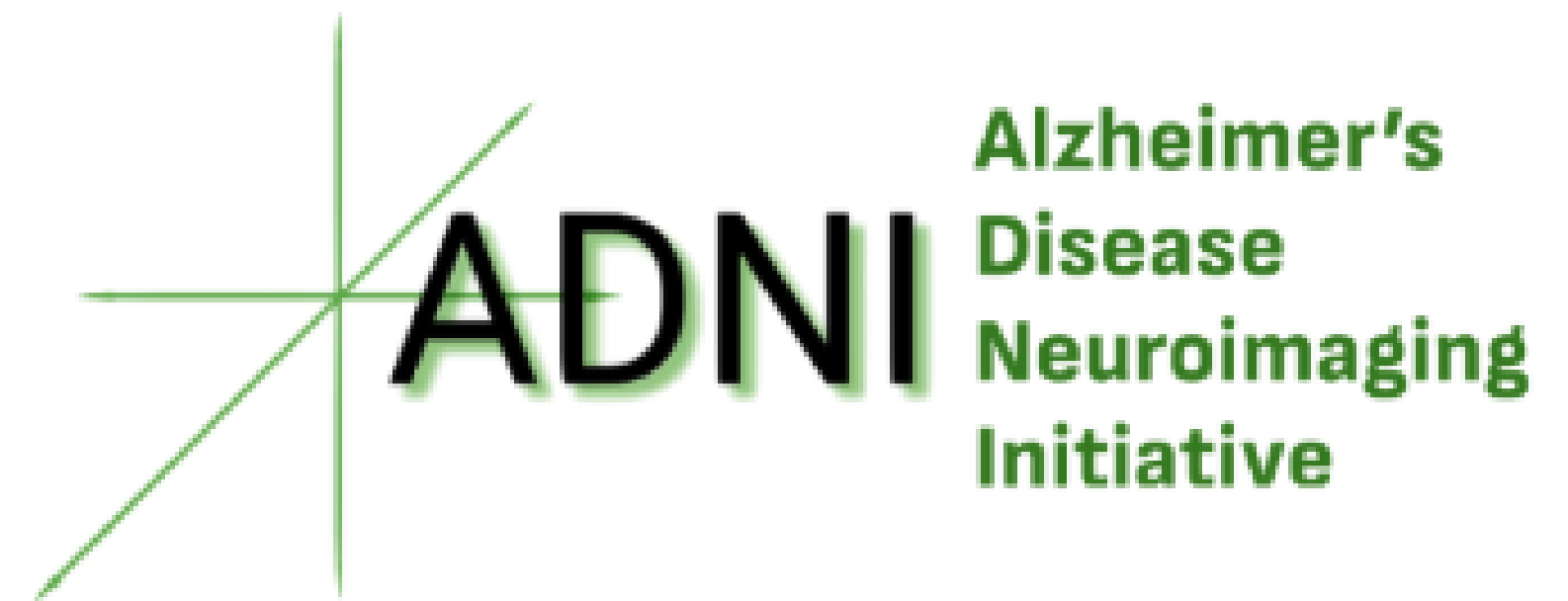
The goal of the project is to segment White Matter Hyperintensities (WMH) which are regions of increased signal intensity on MRI scans, often linked to neurological conditions like small vessel disease, aging-related cognitive decline, and hypertension. The data used consists on FLAIR (Fluid-Attenuated Inversion Recovery) MRI Images. FLAIR is particularly suited for the scope of the project as it's sensitive to pathological brain changes, effective for identifying WMH.



DATA SOURCE

The data were taken from the Alzheimer's Disease Neuroimaging Initiative (ADNI), which is a longitudinal, multi-center, observational study.

The overall goal of ADNI is to validate biomarkers for Alzheimer's disease (AD) clinical trials.



<https://adni.loni.usc.edu/>

CONCEPTUAL FRAMEWORK

Hypothesis

Computing segmentation
relevant
feature maps on the 2D
FLAIR images can improve
Mamdani FIS capabilities

2D feature computation

Research and compute feature maps
that could find small structures in the
scans

Feature Discrimination Capabilities Analysis

How good are the features at
distinguishing the two classes?

Mamdani FIS as binary classifier

Use fuzzy if-then rules and linguistic
variables to predict the class of each pixel

Genetic Algo Optimization

Find the optimal rule weights and
classification thresholds

FEATURE COMPUTATION

For each pixel, a sliding window is centered around it and are computed:

Provided features	Interpretation	Use in the task
Mean	It measures how bright the local neighborhood is. Since WMH appear as hyperintense regions in FLAIR images, high local mean is strong evidence for possible lesion tissue.	Used as a primary intensity feature. A rule or NN can learn that pixels inside bright local areas are more likely to be WMH.
Std	It measures local heterogeneity. A low value means a homogeneous region; a high value indicates intensity variation, often near lesion borders, vessels, or noisy structures.	It helps distinguish uniformly bright tissue from irregular bright regions. It can support lesion detection but can also detect edges or artifacts, so it is usually weaker than mean or top-hat.

Provided features	Interpretation	Use in the task
Skewness	It measures asymmetry of the local intensity distribution. A positive skew means that a few pixels are much brighter than the rest of the local neighborhood.	Useful for detecting small bright abnormalities embedded in darker tissue. In WMH segmentation, it can indicate that a bright lesion-like component is present in the local window, even if the whole region is not uniformly bright.
Kurtosis	It measures how peaked or heavy-tailed the local intensity distribution is. High kurtosis means that the window contains extreme intensity values compared to the local background.	It can highlight small, sharp hyperintense structures. However, it may also respond to noise or isolated artifacts, so it is usually complementary rather than decisive.

ADDITIONAL FEATURE COMPUTATION

Feature	Interpretation	Use in the task
Local Z-score	Measures how much brighter or darker each pixel is compared to its local neighborhood. A high positive local z-score means that the pixel is unusually bright compared to nearby pixels.	This is useful for WMH segmentation because very small WMH may not strongly affect the local mean, but they can still appear as locally abnormal bright spots
White top-hat filter	Enhances small bright structures in the image.	This makes it suitable for detecting small WMH regions that may otherwise be lost in broader intensity statistics. The radius of the structuring element controls the lesion scale being emphasized: smaller radii detect smaller spots, while larger radii respond to larger regions.

WHITE TOP HAT FILTER

The idea is to enhance small bright objects

TopHat = Original image – Opened image



Erosion → Dilation

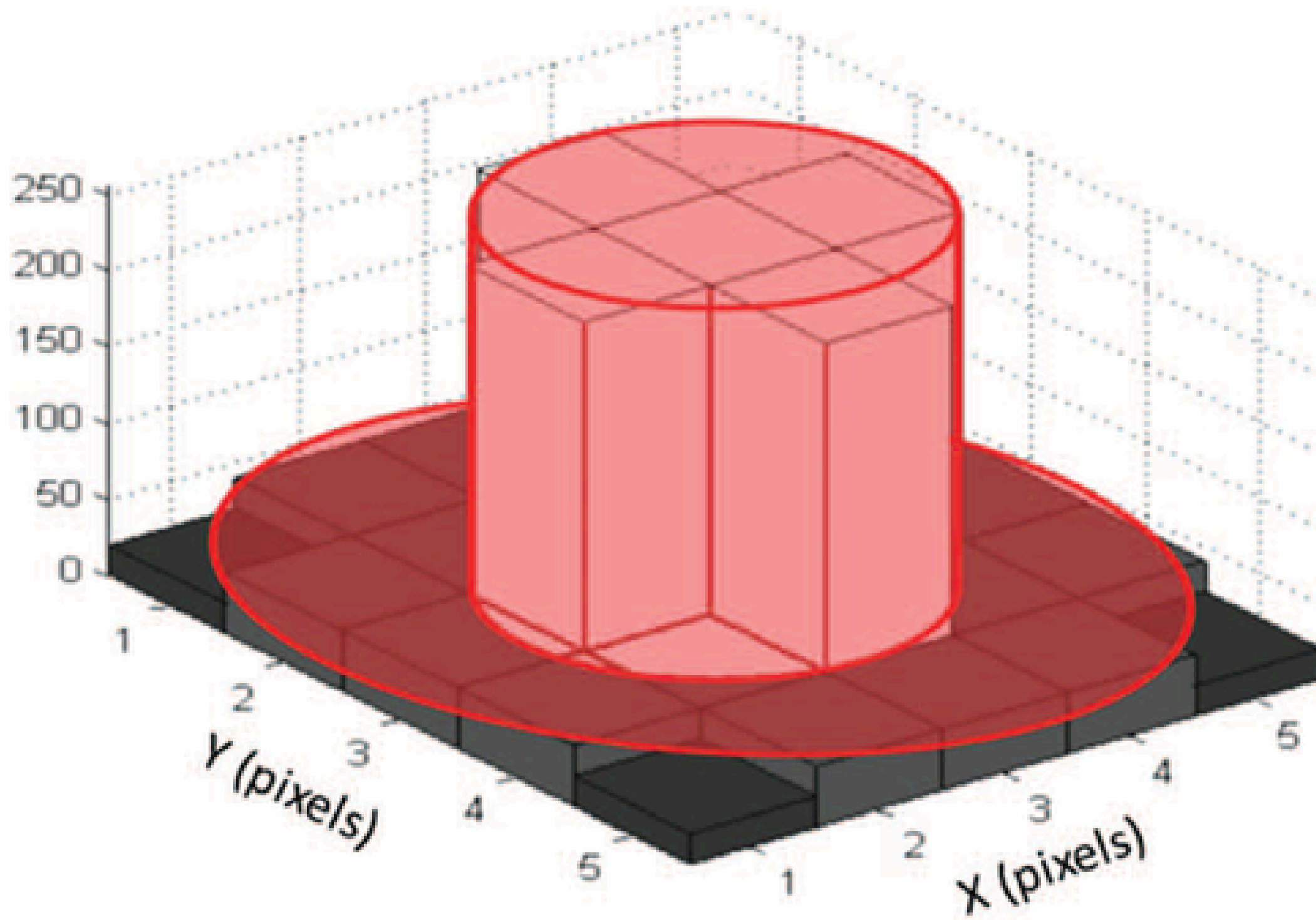


For each pixel, checks whether the whole structuring element fits inside the bright object. If it does not fit, the pixel is removed.

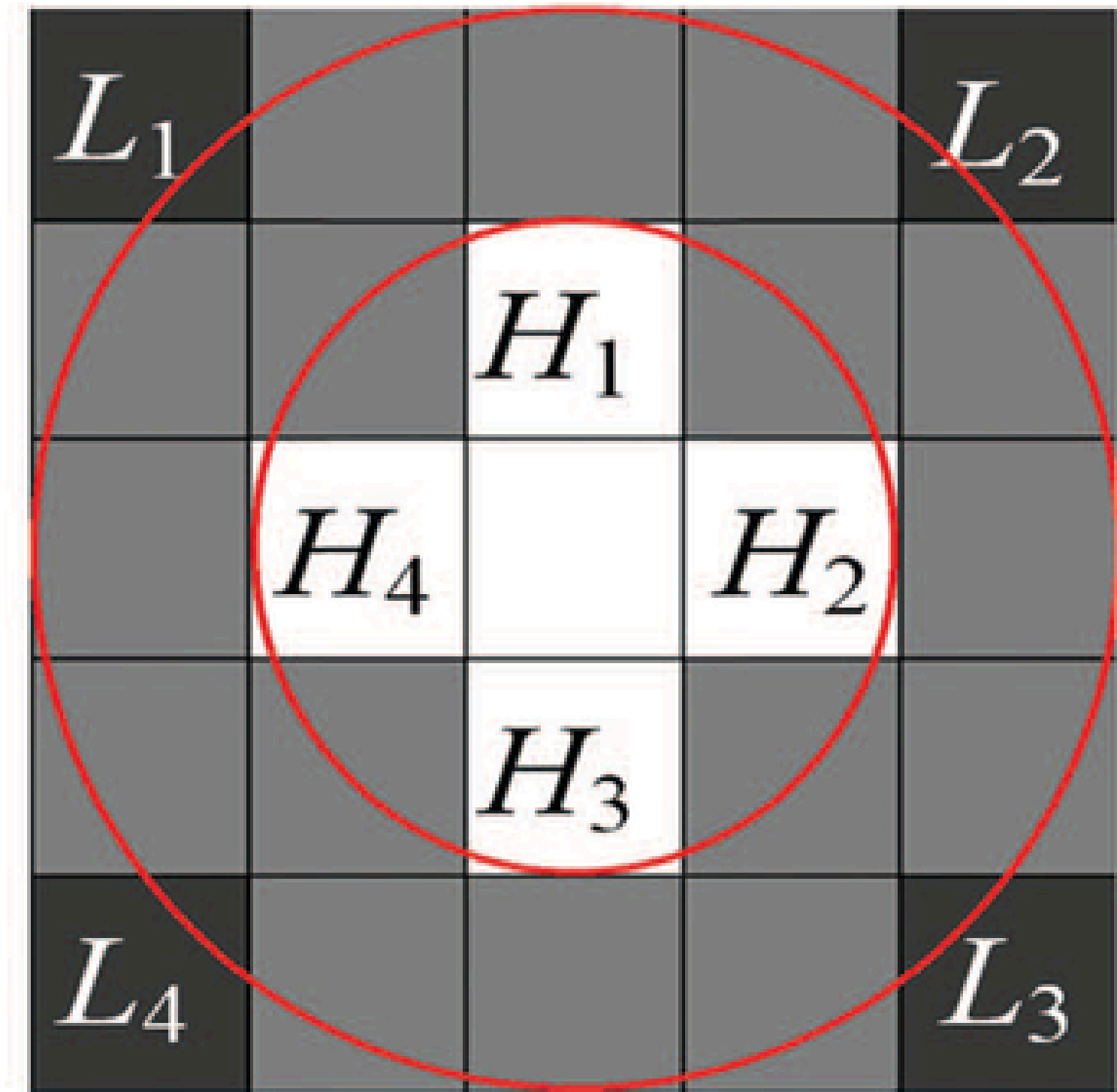


checks nearby bright pixels and spreads them outward.

Removes small bright objects.



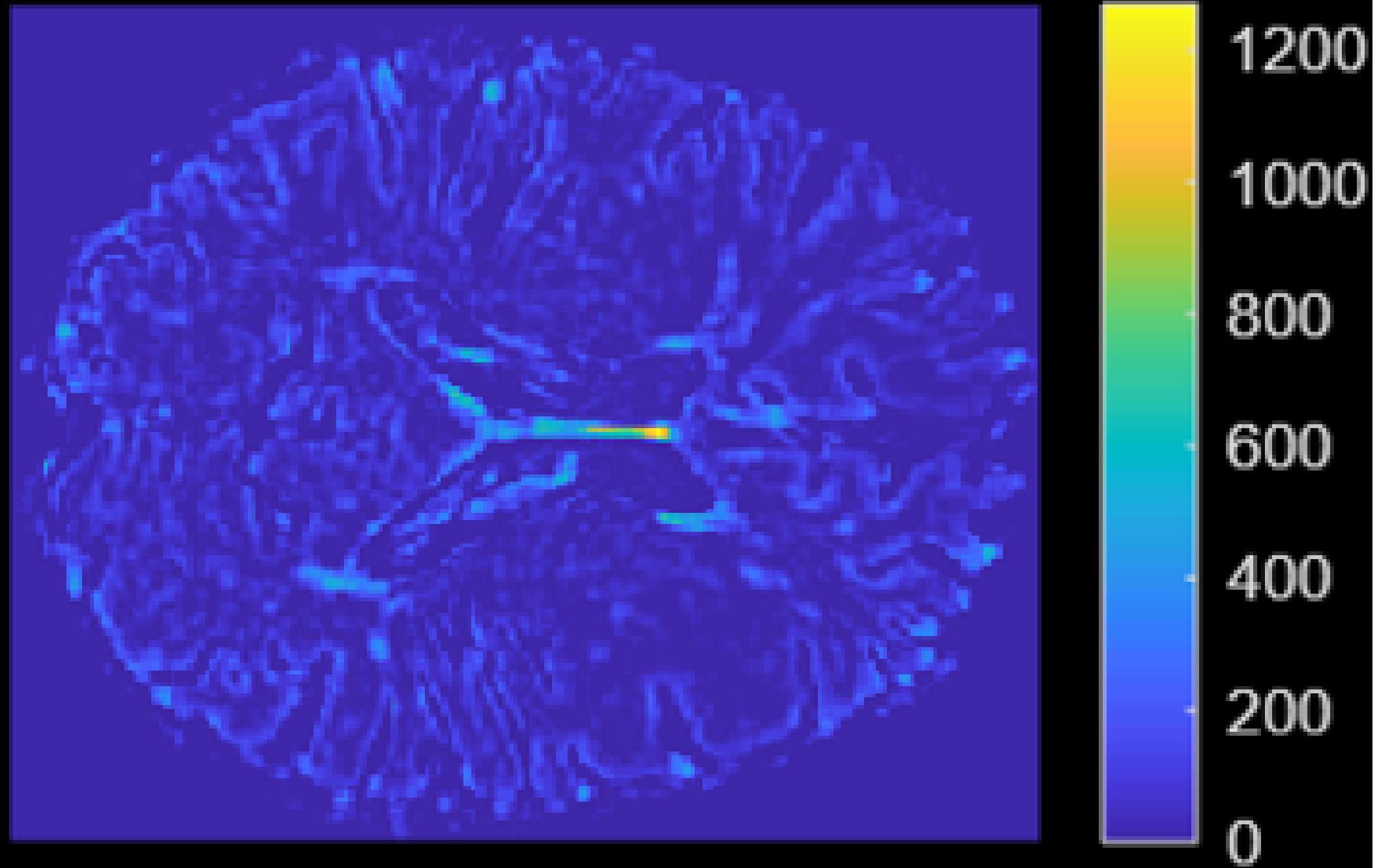
(a)



(b)

TY - JOUR; AU - Chang, Ruey-Feng; AU - Hou, Yu-Ling; AU - Huang, C. F.; AU - Chen, Jeon-Hor; AU - Chang, Jung; AU - Moon, Woo; Kyung; PY - 2013/09/17; SP - 102901; T1 - **Automatic detection of microcalcifications in breast ultrasound**; VL - 40; DO - 10.1118/1.4821098; JO - Medical Physics

White top-hat



ADDITIONAL FEATURE COMPUTATION

Feature	Interpretation	Use in the task
Local Prior	We want to answer the question “Spatial plausibility: is this pixel located in a region where WMH are usually observed?”	The utility comes from the fact that very bright regions might be found in areas of the brain where it is not probable to find the WMH, thus, through the fuzzy rules, they can be ignored
Regional prior	“Is this pixel close to one of the main spatial regions where WMH commonly appear?”	Local prior can be too strict so compute a gaussian filter to include areas around the main found priors
Centrality	“Is the pixel close to the center of the brain or further away?”	After some EDA one can see that there are very bright areas far away from the brain center which are not WMH and they can be misleading

STRATIFICATION

The aim of this step is to have a representative sample for the computation of the spatial prior.

In particular the script:

1. finds all diagnostic groups present in the dataset,
2. compute how many patients each class has
3. compute the ideal 80% quota for each class (ideal quota = $0.80 \times$ number of patients in that class)
4. it gives the remaining patients to the classes with the largest decimal remainder: this keeps the final split as close as possible to the original diagnosis distribution.
5. randomly select patients inside each class,
6. define the held-out patients

```
Spatial prior will be computed using 489 / 611 images (80.0%).
```

```
Diagnosis distribution in full dataset:
```

```
Diagnosis 0: 527 / 611 images (86.25%)
```

```
Diagnosis 1: 84 / 611 images (13.75%)
```

```
Diagnosis distribution in prior subset:
```

```
Diagnosis 0: 422 / 489 images (86.30%)
```

```
Diagnosis 1: 67 / 489 images (13.70%)
```


LOCAL PRIOR

It is a 2D probability map: “Is this pixel located in a region where WMH are usually observed?”

Computed using only the stratified 80% prior-training subset.

For each pixel:

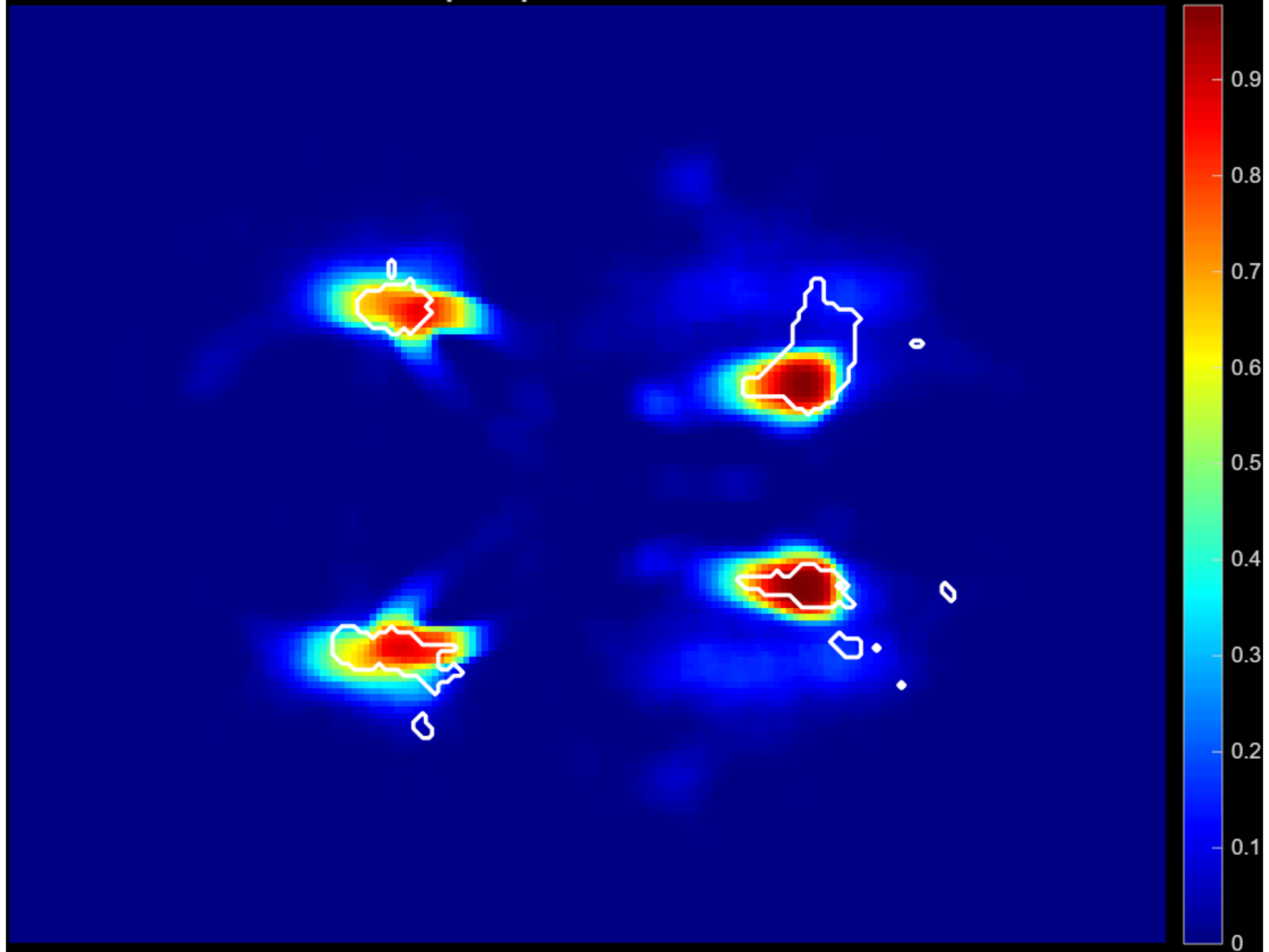
- look at a 5×5 window around that pixel,

for each patient used to build the prior, extract the WMH mask:

- count how many WMH pixels exist inside the 5×5 neighborhood. If the count is greater than zero, that pixel is marked as true.

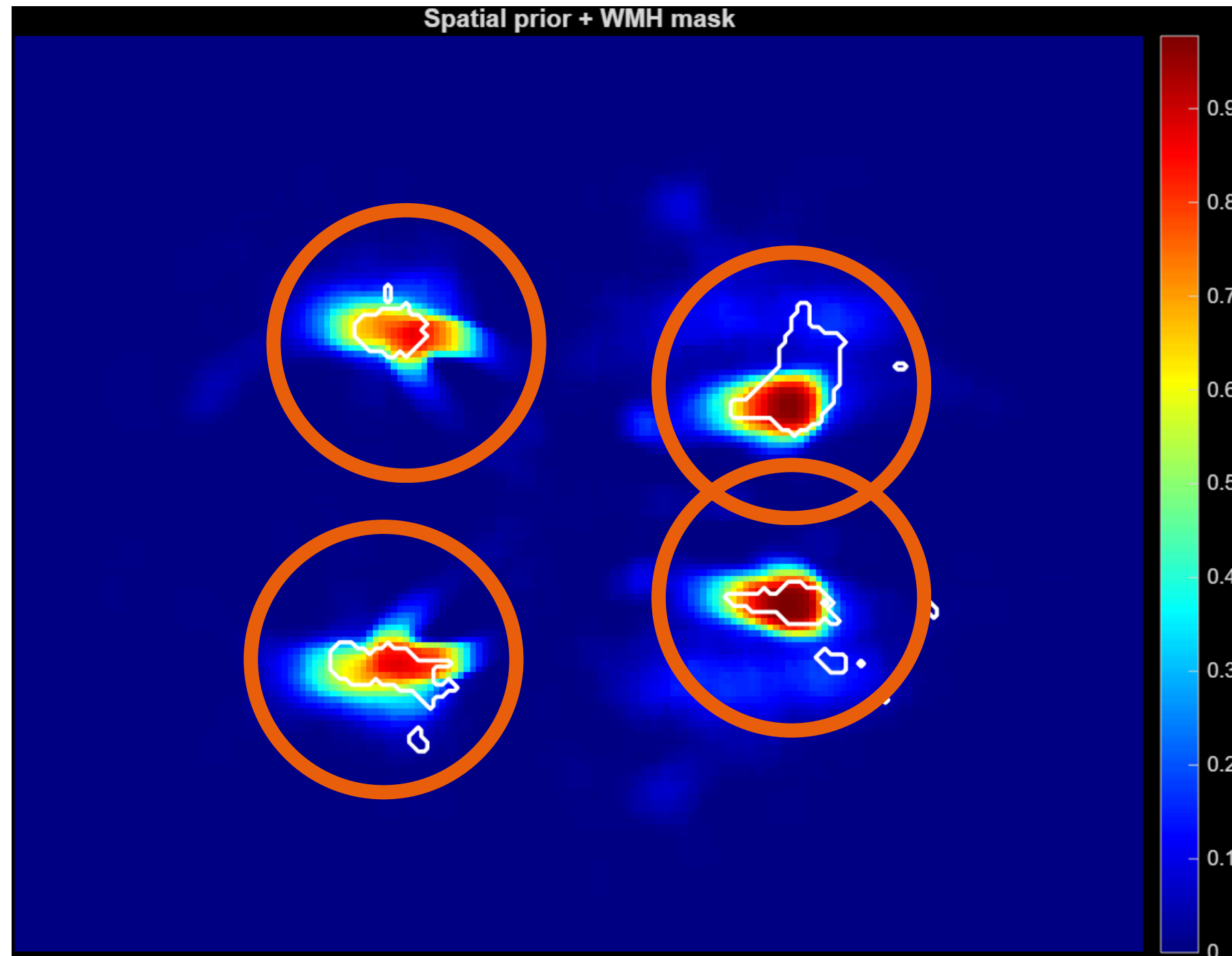
After the this unnormalized prior is computed for each selected patients, the prior is averaged over all the patients so that at the end each pixel intensity represents: “In x% of the prior-training patients, location (i,j) had at least one WMH pixel in its 5×5 neighborhood.”

Spatial prior + WMH mask



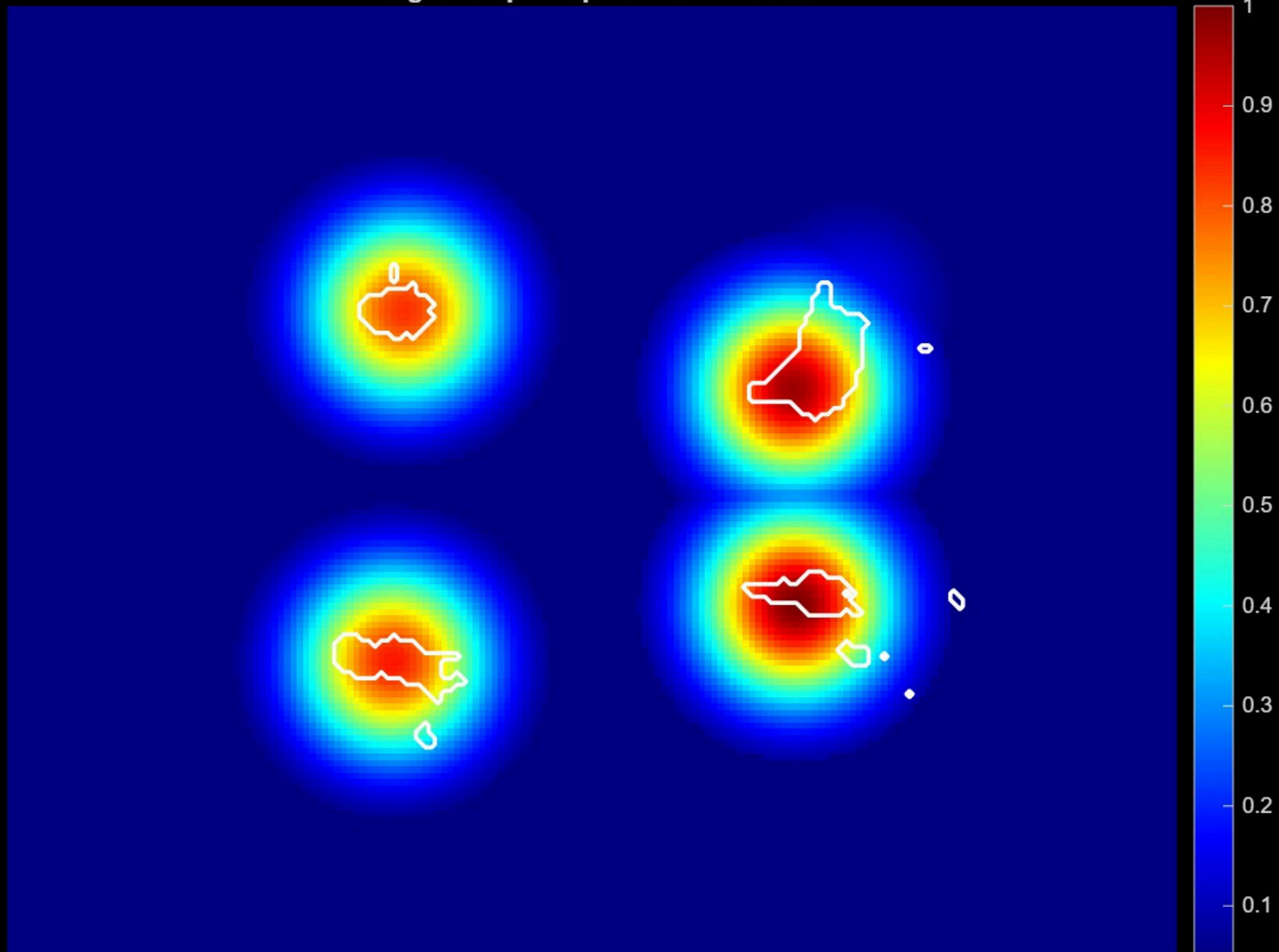
REGIONAL PRIOR

Is this pixel close to a frequent WMH region?



1. Select the brightest prior regions
2. Compute prior value weighted regions centers
 - a. each pixel in the brightest regions pulls the center coordinates with a strength proportional to its brightness
3. Create a Gaussian region around each center
 - a. the distance from the peak center is computed
 - b. keep only pixels within a circular region around the centers
 - c. Inside that circular region, create a soft Gaussian heat map

Regional spatial prior + WMH mask



FEATURE SIGNIFICANCE ANALYSIS

This is used to evaluate which features are useful for distinguishing WMH pixels from non-WMH pixels. In other words, before designing the FIS rules, it asks: "For each feature, are WMH pixels statistically different from background brain pixels?"

Metric	Interpretation	Use in the task
Mean inside WMH vs mean outside WMH	This compares the average feature value in the lesion area against the average feature value in normal brain tissue.	It tells the “direction” of the feature. For instance: if Mean WMH = 0.75 and Mean non-WMH = 0.32. This means WMH pixels tend to have higher values for that feature
Cohen's d (effect size)	This measures how separated the two groups are, taking variability into account (measured in stds). This is introduced because just comparing means is not enough.	Cohen himself suggested this interpretation of the values: d around 0.2 → weak separation d around 0.5 → moderate separation d above 0.8 → strong separation

Metric	Interpretation	Use in the task
ROC AUC	“Do WMH pixels generally receive higher feature values than non-WMH pixels?”. In the computation of this metric it doesn't matter that the data are coming from a 2D image, this is because it is assessing if the feature value is generally different for instances of the two classes, so if a threshold can be found to separate them.	We can tell if in the feature map provided the pixels of the class are generally assigned different values or very similar ones, so tell how much in general the considered feature is different per classes.

ROC AUC $\approx 1 \rightarrow$ WMH pixels usually have higher feature values than non-WMH pixels

ROC AUC $\approx 0.5 \rightarrow$ the feature values for the two classes are similar so it is not very discriminative

ROC AUC $\approx 0 \rightarrow$ the feature is separable but in the opposite direction: WMH pixels usually have lower values.

FEATURE SIGNIFICANCE ANALYSIS

===== Average feature significance over held-out 20% prior-test set =====

Feature: Mean		Feature: Skew		Feature: LocalZ	
Mean WMH:	119077.3022	Mean WMH:	-0.1964	Mean WMH:	0.6553
Mean non-WMH:	76178.2198	Mean non-WMH:	-0.0102	Mean non-WMH:	-0.0081
Cohen d:	1.6013	Cohen d:	-0.2549	Cohen d:	1.0530
Abs Cohen d:	1.6013	Abs Cohen d:	0.2790	Abs Cohen d:	1.0530
AUC:	0.9109	AUC:	0.4289	AUC:	0.7670
AUC abs:	0.9109	AUC abs:	0.5786	AUC abs:	0.7670
Feature: Std		Feature: Kurtosis		Feature: TopHat	
Mean WMH:	12021.4440	Mean WMH:	-0.5194	Mean WMH:	13316.3205
Mean non-WMH:	11072.0141	Mean non-WMH:	-0.2319	Mean non-WMH:	6020.8148
Cohen d:	0.3623	Cohen d:	-0.2588	Cohen d:	1.1674
Abs Cohen d:	0.4206	Abs Cohen d:	0.2766	Abs Cohen d:	1.1674
AUC:	0.6595	AUC:	0.4322	AUC:	0.8207
AUC abs:	0.6657	AUC abs:	0.5749	AUC abs:	0.8207

FEATURE SIGNIFICANCE ANALYSIS

Feature: Distance center	Feature: Regional spatial prior
Mean WMH: 0.4339	Mean WMH: 0.7136
Mean non-WMH: 0.5925	Mean non-WMH: 0.1192
Cohen d: -0.9867	Cohen d: 2.8314
Abs Cohen d: 0.9867	Abs Cohen d: 2.8314
AUC: 0.2510	AUC: 0.9329
AUC abs: 0.7490	AUC abs: 0.9329
Feature: Spatial prior	
Mean WMH: 0.6175	
Mean non-WMH: 0.0373	
Cohen d: 2.8611	
Abs Cohen d: 2.8611	
AUC: 0.9651	
AUC abs: 0.9651	

FEATURE SIGNIFICANCE ANALYSIS

==== Brain-relative spatial interpretation ====

X position:

Mean X WMH: 0.5370

Mean X non-WMH: 0.4924

Y position:

Mean Y WMH: 0.5010

Mean Y non-WMH: 0.4997

Distance from brain center:

Mean Dist WMH: 0.4440

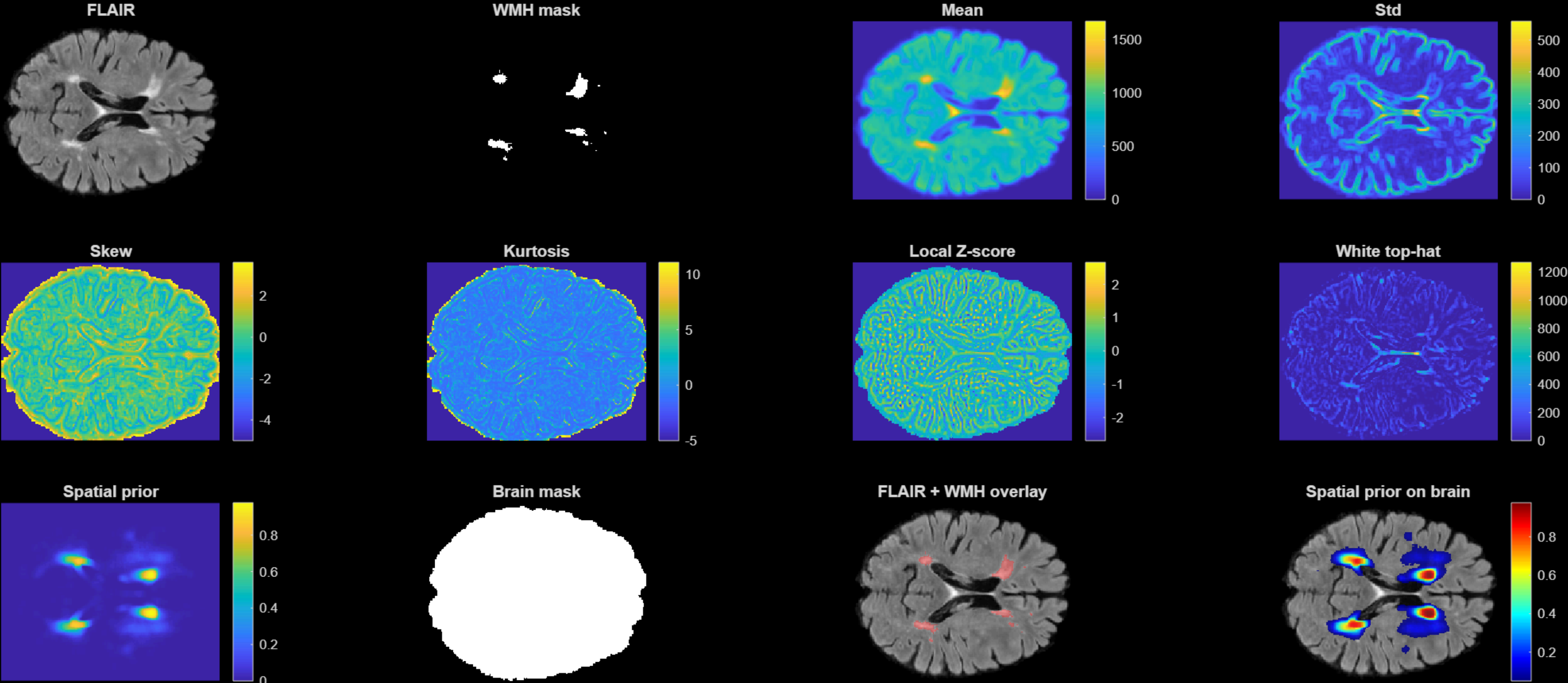
Mean Dist non-WMH: 0.5926

AUC X position:	0.5558		AUC abs: 0.5558
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AUC Y position:	0.5018		AUC abs: 0.5018
-----------------	--------	--	-----------------

AUC distance center:	0.2631		AUC abs: 0.7369
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FEATURE VISUALIZATION



PREPROCESSING

Input

- 1. FLAIR_dataset.npy
- 2. WMH_masks.npy
- 3. mean_dataset.npy
- 4. std_dataset.npy
- 5. skew_dataset.npy
- 6. kurtosis_dataset.npy
- cached_extra_features.mat

Output

preprocessed_features_for_fis
.mat

Creation of the brain masks

All the pixels outside the mask are set to be zero and those values are not used in the subsequent clipping and normalization phases.

Nan Inf removal

Every feature is converted into double data type along with the removal of problematic value such as Nan/Inf

Values Clipping

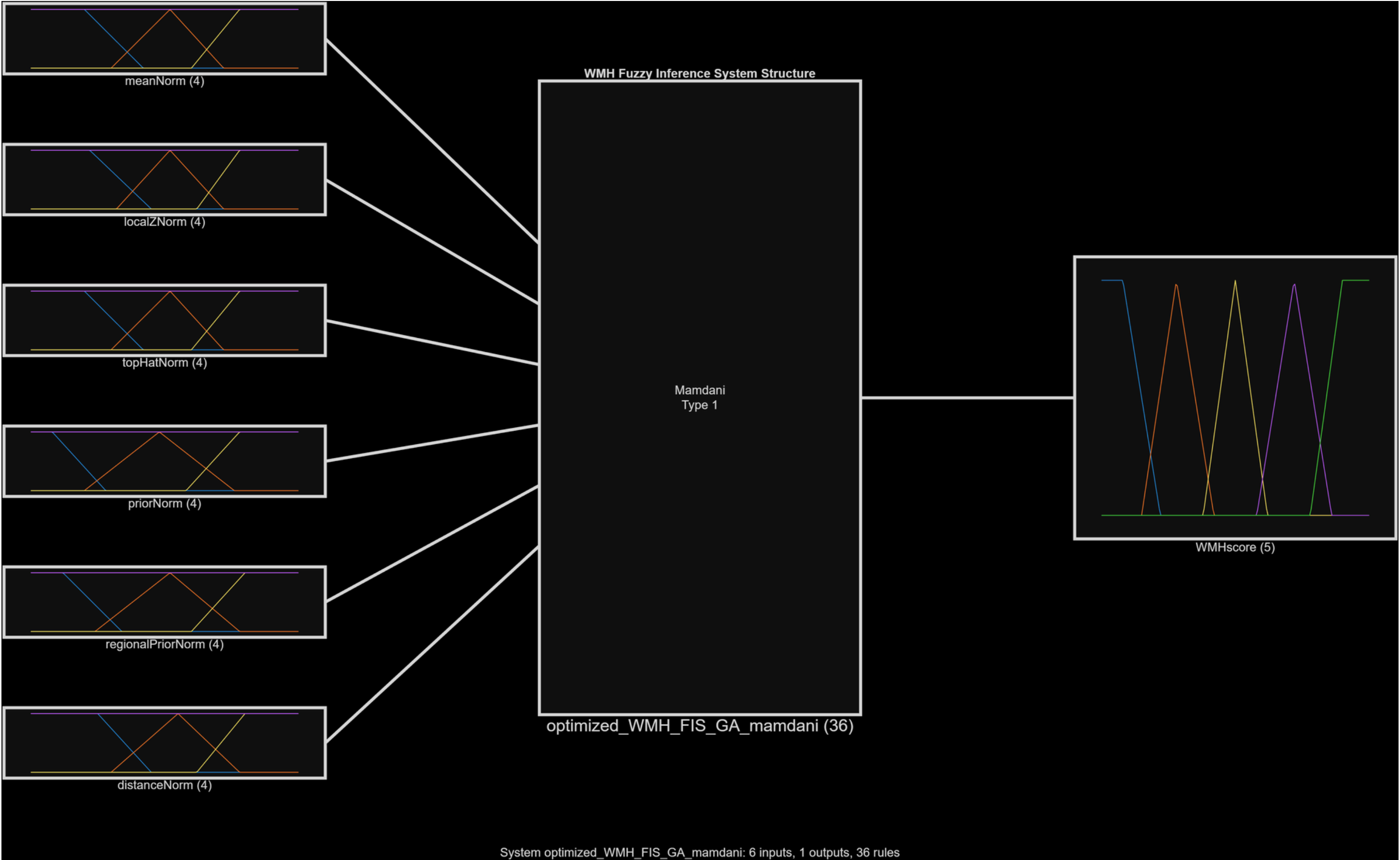
Since features such as localZFeat, topHatFeat, skewFeat might have very high values using a min-max clipping would squash small values very close to zero, making them hardly recognizable. Clip very low and high values to the percentiles,

Normalization in [0,1]

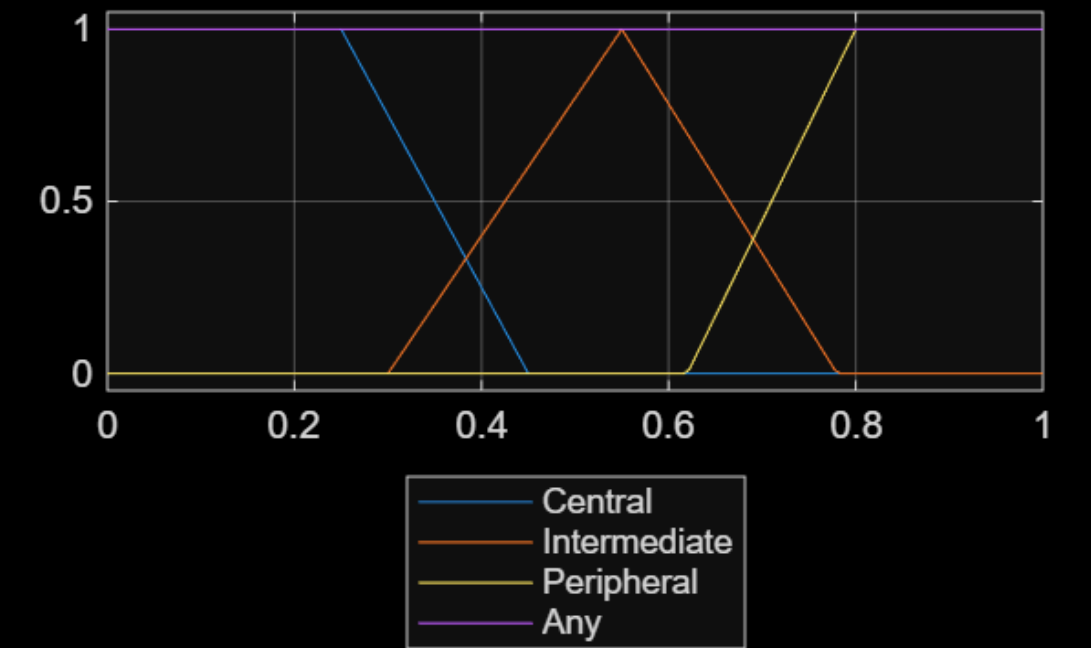
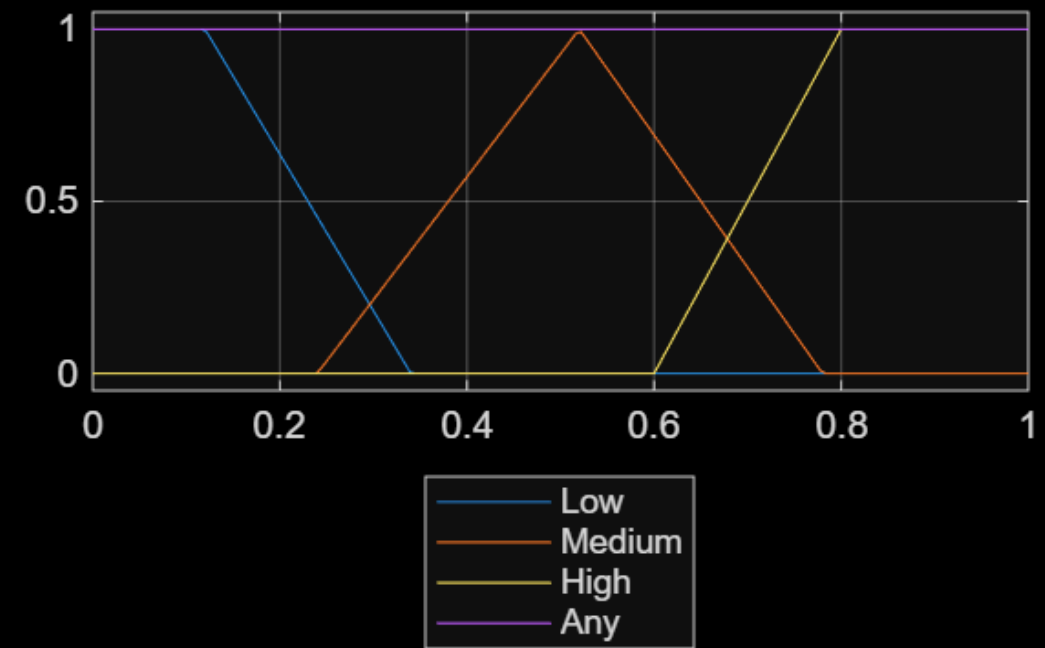
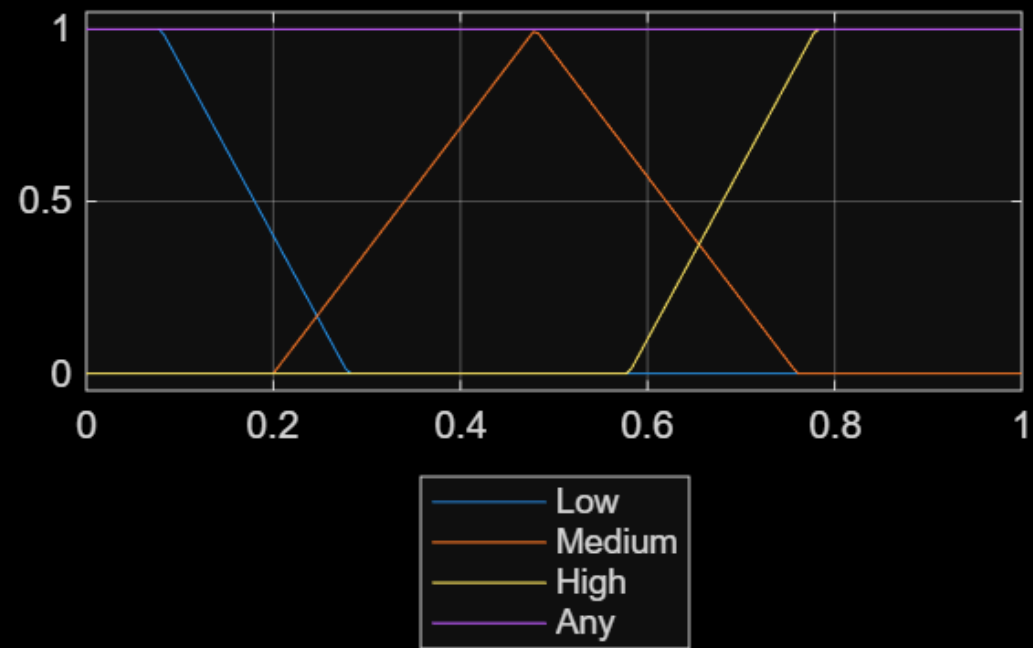
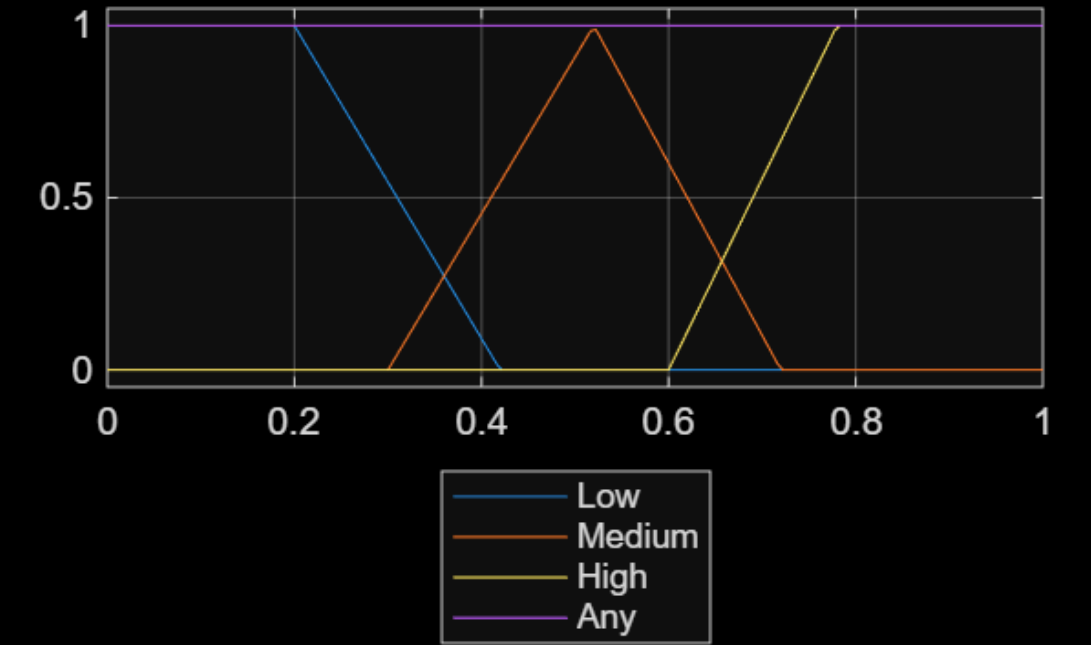
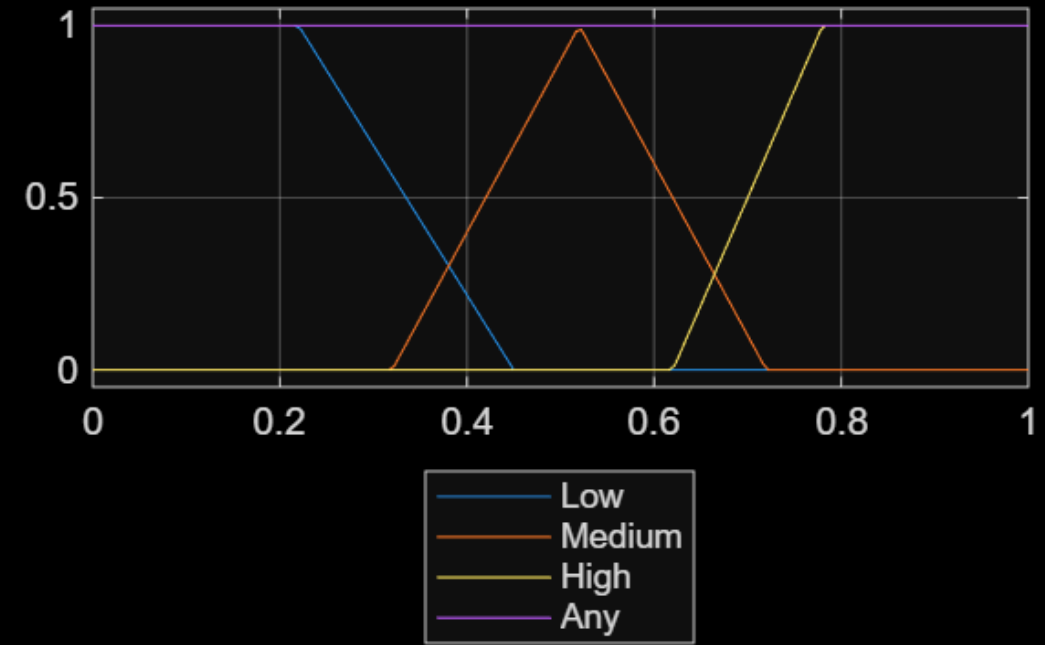
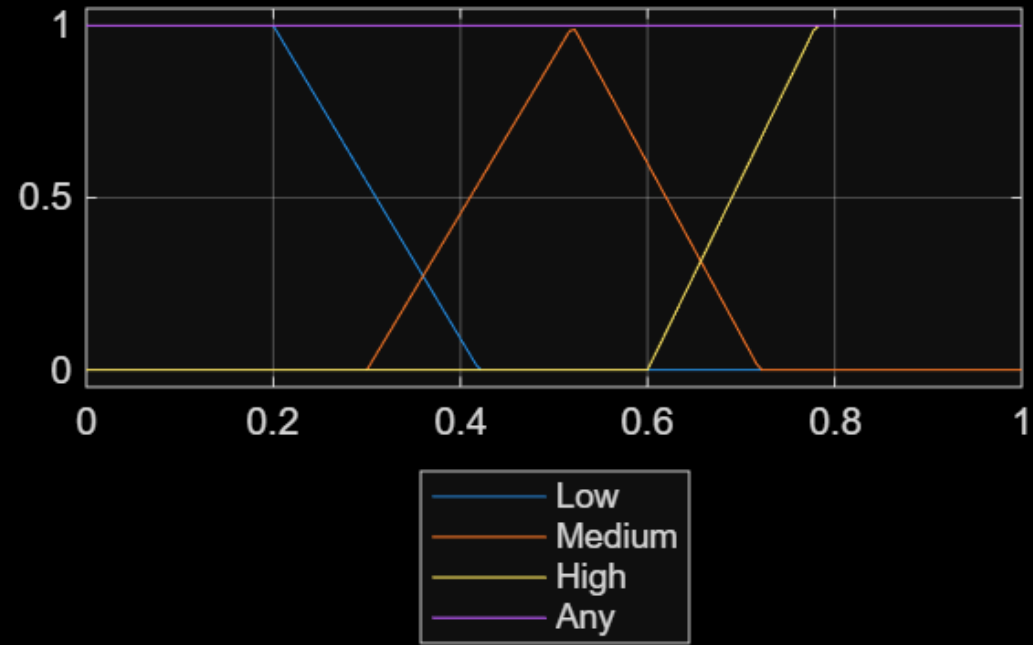
MAMDANI FIS

- 1 ● Create Mamdani FIS
- 2 ● Define input features, membership functions, output classes, and fuzzy rules
- 3 ● Call a genetic algorithm to optimize the rule weights

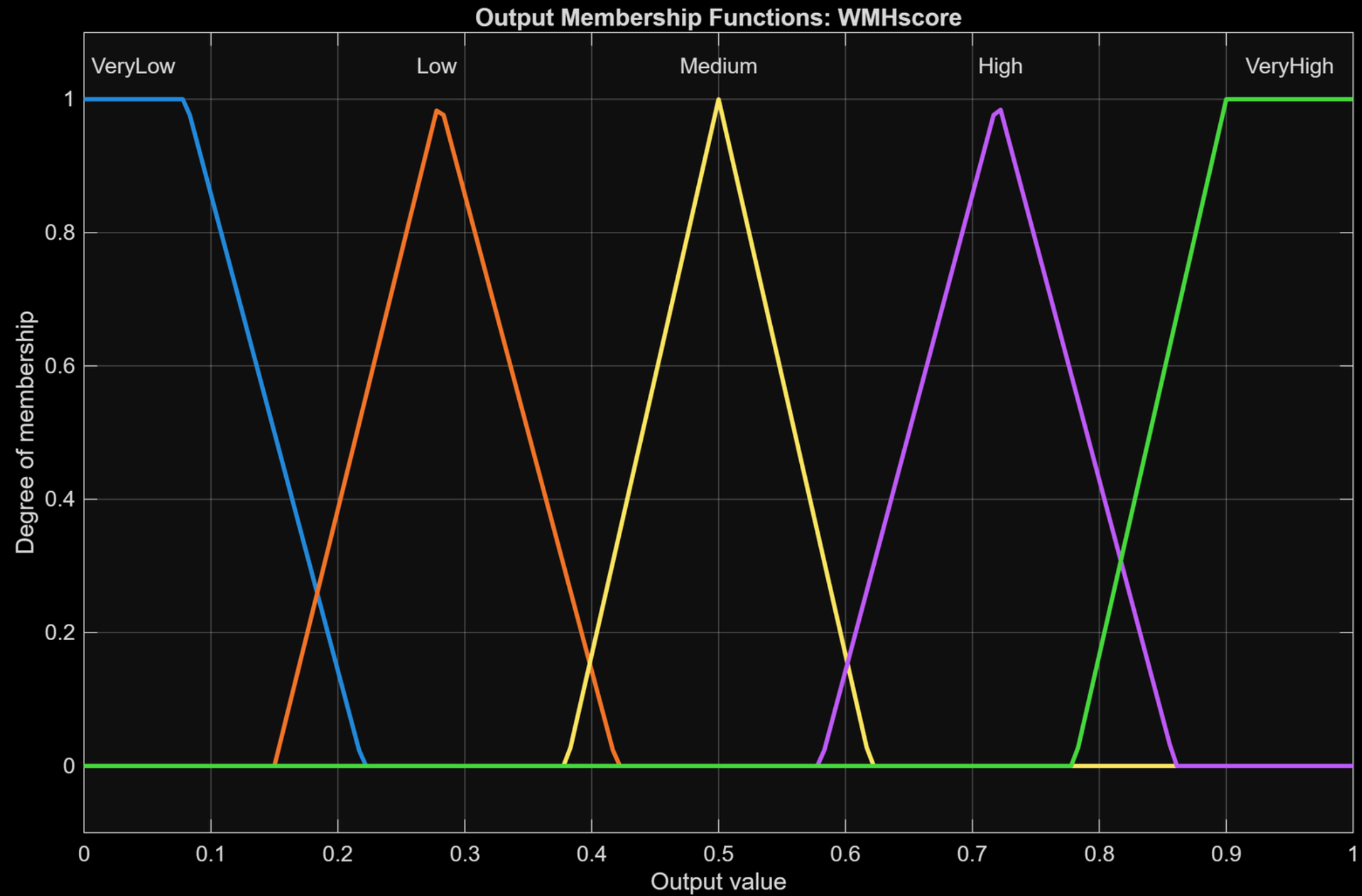
MAMDANI FIS



MAMDANI FIS -INPUT FEATURES



MAMDANI FIS -OUTPUT



MAMDANI FIS -RULES

Rule	RuleText
1	ans = '1. If (meanNorm is High) and (localZNorm is High) and (topHatNorm is High) then (WMHscore is VeryHigh) (0.97239)'
2	ans = '2. If (meanNorm is High) and (localZNorm is High) and (regionalPriorNorm is High) then (WMHscore is VeryHigh) (1)'
3	ans = '3. If (meanNorm is High) and (topHatNorm is High) and (regionalPriorNorm is High) then (WMHscore is VeryHigh) (1)'
4	ans = '4. If (meanNorm is High) and (localZNorm is High) and (priorNorm is High) then (WMHscore is VeryHigh) (1)'
5	ans = '5. If (meanNorm is High) and (topHatNorm is High) and (priorNorm is High) then (WMHscore is VeryHigh) (1)'
6	ans = '6. If (meanNorm is High) and (localZNorm is Medium) and (topHatNorm is High) and (regionalPriorNorm is High) then (WMHscore is High) (0.90242)'
7	ans = '7. If (meanNorm is Medium) and (localZNorm is High) and (topHatNorm is High) and (regionalPriorNorm is High) then (WMHscore is High) (0.99362)'
8	ans = '8. If (meanNorm is High) and (localZNorm is High) and (topHatNorm is Medium) and (regionalPriorNorm is Medium) then (WMHscore is High) (1)'
Rule	RuleText
9	ans = '9. If (meanNorm is High) and (localZNorm is Medium) and (topHatNorm is Medium) and (priorNorm is High) and (regionalPriorNorm is High) then (WMHscore is High) (0.82926)'
10	ans = '10. If (meanNorm is High) and (priorNorm is High) and (regionalPriorNorm is High) and (distanceNorm is Central) then (WMHscore is High) (1)'
11	ans = '11. If (meanNorm is Medium) and (localZNorm is High) and (priorNorm is High) and (regionalPriorNorm is High) then (WMHscore is High) (1)'
12	ans = '12. If (localZNorm is High) and (topHatNorm is High) and (priorNorm is Low) and (regionalPriorNorm is Low) then (WMHscore is Medium) (1)'
13	ans = '13. If (meanNorm is Low) then (WMHscore is VeryLow) (1)'
14	ans = '14. If (localZNorm is Low) and (topHatNorm is Low) then (WMHscore is VeryLow) (1)'
15	ans = '15. If (meanNorm is Medium) and (localZNorm is Low) and (topHatNorm is Low) then (WMHscore is VeryLow) (1)'
16	ans = '16. If (meanNorm is High) and (localZNorm is Low) and (topHatNorm is Low) then (WMHscore is Low) (1)'

MAMDANI FIS -RULES

Rule	RuleText
17	ans = '17. If (localZNorm is Medium) and (topHatNorm is Low) and (priorNorm is Low) and (regionalPriorNorm is Low) and (distanceNorm is Peripheral) then (WMHscore is VeryLow) (0.97116)'
18	ans = '18. If (meanNorm is Any) and (localZNorm is Any) and (topHatNorm is Any) and (priorNorm is Any) and (regionalPriorNorm is Any) and (distanceNorm is Any) then (WMHscore is VeryLow) (0.02)'
19	ans = '19. If (meanNorm is High) and (localZNorm is High) and (topHatNorm is High) then (WMHscore is VeryHigh) (0.79308)'
20	ans = '20. If (meanNorm is High) and (localZNorm is High) and (regionalPriorNorm is High) then (WMHscore is VeryHigh) (1)'
21	ans = '21. If (meanNorm is High) and (topHatNorm is High) and (regionalPriorNorm is High) then (WMHscore is VeryHigh) (0.97715)'
22	ans = '22. If (meanNorm is High) and (localZNorm is High) and (priorNorm is High) then (WMHscore is VeryHigh) (1)'
23	ans = '23. If (meanNorm is High) and (topHatNorm is High) and (priorNorm is High) then (WMHscore is VeryHigh) (1)'
24	ans = '24. If (meanNorm is High) and (localZNorm is Medium) and (topHatNorm is High) and (regionalPriorNorm is High) then (WMHscore is High) (1)'
Rule	RuleText
25	ans = '25. If (meanNorm is Medium) and (localZNorm is High) and (topHatNorm is High) and (regionalPriorNorm is High) then (WMHscore is High) (0.96313)'
26	ans = '26. If (meanNorm is High) and (localZNorm is High) and (topHatNorm is Medium) and (regionalPriorNorm is Medium) then (WMHscore is High) (1)'
27	ans = '27. If (meanNorm is High) and (localZNorm is Medium) and (topHatNorm is Medium) and (priorNorm is High) and (regionalPriorNorm is High) then (WMHscore is High) (1)'
28	ans = '28. If (meanNorm is High) and (priorNorm is High) and (regionalPriorNorm is High) and (distanceNorm is Central) then (WMHscore is High) (1)'
29	ans = '29. If (meanNorm is Medium) and (localZNorm is High) and (priorNorm is High) and (regionalPriorNorm is High) then (WMHscore is High) (1)'
30	ans = '30. If (localZNorm is High) and (topHatNorm is High) and (priorNorm is Low) and (regionalPriorNorm is Low) then (WMHscore is Medium) (1)'
31	ans = '31. If (meanNorm is Low) then (WMHscore is VeryLow) (1)'
32	ans = '32. If (localZNorm is Low) and (topHatNorm is Low) then (WMHscore is VeryLow) (1)'

MAMDANI FIS -RULES

Rule	RuleText
33	ans = '33. If (meanNorm is Medium) and (localZNorm is Low) and (topHatNorm is Low) then (WMHscore is VeryLow) (1)'
34	ans = '34. If (meanNorm is High) and (localZNorm is Low) and (topHatNorm is Low) then (WMHscore is Low) (1)'
35	ans = '35. If (localZNorm is Medium) and (topHatNorm is Low) and (priorNorm is Low) and (regionalPriorNorm is Low) and (distanceNorm is Peripheral) then (WMHscore is VeryLow) (1)'
36	ans = '36. If (meanNorm is Any) and (localZNorm is Any) and (topHatNorm is Any) and (priorNorm is Any) and (regionalPriorNorm is Any) and (distanceNorm is Any) then (WMHscore is VeryLow) (0.03)'

GENETIC OPTIMIZATION

1 ● Patient Selection

- The script selects the specified number of patients from **priorTrainIdx**.
- It counts how many WMH pixels each patient has,
- Then it keeps only WMH-positive patients and selects the ones with the largest lesion load,

Rationale: patients with visible lesions provide more informative training examples. If patients with almost no WMH were used, the GA could learn to predict almost everything as non-WMH and still obtain apparently good accuracy.

2 Chromosome Representation

Each GA individual is a vector:

[ruleWeight_1 ... ruleWeight_N threshold]

The rule weights are constrained in [0,1] while the threshold in [0.45,0.85]

3 Population Initialization

- The initial population is not fully random. It starts from the handcrafted FIS
- The remaining individuals are random combinations of rule weights and threshold values.

4 ● Fitness Evaluation

- For each individual, the script creates a candidate FIS,
- Then it evaluates the FIS on all selected training patients:
- The score is converted into a binary WMH prediction using the threshold

5 ● Segmentation Metrics

The script computes the main pixel-wise segmentation metrics:

- $\text{Dice} = 2TP / (2TP + FP + FN)$
- $\text{Sensitivity} = TP / (TP + FN)$

Measures overlap between prediction and ground truth

Measures how many real WMH pixels are detected

6 Segmentation Metrics

- Precision = $TP / (TP + FP)$
- Specificity = $TN / (TN + FP)$
- FPR = $FP / (FP + TN)$
- FNR = $FN / (FN + TP)$

Measures how many predicted WMH pixels are actually correct.

Measures how well non-WMH pixels are rejected.

FPR measures false alarms. FNR measures missed lesions.

7 Loss Function

```
loss = opts.lambdaDice * (1 - metrics.Dice) ...  
      + opts.lambdaFNR * metrics.FNR ...  
      + opts.lambdaFPR * metrics.FPR ...  
+ opts.lambdaSens * (1 - metrics.Sensitivity) ...  
+ opts.lambdaPrec * (1 - metrics.Precision);
```

The main objective is to maximize Dice, but the loss also penalizes:

- **high false negative rate;**
- **high false positive rate;**
- **low sensitivity;**
- **low precision.**

8 ● Loss Function

Two extra penalties are added:
if metrics.Sensitivity < 0.35

 loss = loss + 1.5;

end

if metrics.Dice < 0.10

 loss = loss + 1.0;

end

These prevent degenerate solutions, especially predictions that miss most lesions or produce almost no meaningful overlap.

9 ● Genetic Algorithm Operations

At each generation:

- the population is sorted by fitness:
- The best individuals are copied directly to the next generation: elitism
- Parents are selected using tournament selection
- New children are created using arithmetic crossover
- Then mutation adds Gaussian noise
- Finally, values are clipped inside the allowed ranges.

NEURAL WMH SEGMENTATION

FEATURE SET USED BY THE NEURAL NETWORK

```
feature_cols = [  
    "mean",  
    "std",  
    "skew",  
    "kurtosis",  
    "localZ",  
    "topHat",  
    "prior",  
    "regionalPrior",  
    "distance",  
    ]
```

The MRI segmentation problem is transformed into a tabular supervised learning problem.

Each row has this structure:

```
patient_id | row | col | features... | WMH label
```

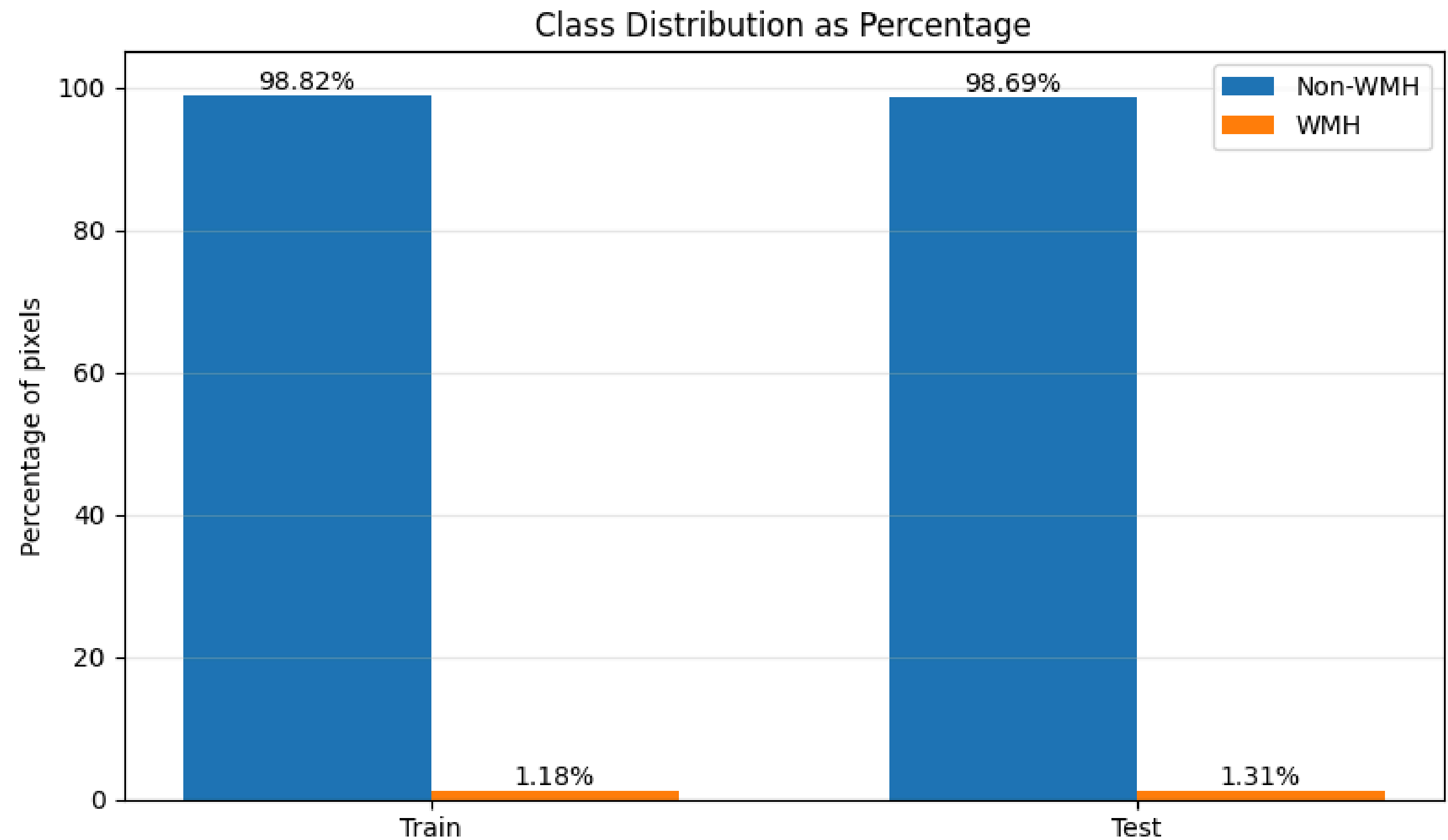
The inclusion of patient_id is essential because it allows later splitting by patient rather than by pixel.

TRAIN/TEST SPLIT USING PRIOR TABLE

The test set contains patients not used for prior construction. This is important because the spatial prior uses lesion masks from training patients, so test patients must remain excluded from that computation to avoid information leakage.

Train test split is also performed using the patient index this is because neighboring pixels from the same MRI image are highly correlated. If random pixel splitting were used, pixels from the same patient could appear in both training and test sets, which again would cause data leakage.

CLASS IMBALANCE ANALYSIS



Class weights

0.5 42.3

$$\text{WEIGHT_C} = \frac{\text{TOTAL SAMPLES}}{(\text{NUMBER OF CLASSES} \times \text{SAMPLES IN CLASS C})}$$

CLASS IMBALANCE ANALYSIS

$$\text{WEIGHT_C} = \text{TOTAL SAMPLES} / (\text{NUMBER OF CLASSES} \times \text{SAMPLES IN CLASS C})$$

Class weights

0.5 42.3

NEURAL NETWORK ARCHITECTURE

Input layer: 9 features



Dense layer: 64 neurons, ReLU



Dropout: 30%



Dense layer: 32 neurons, ReLU



Dropout: 30%



Dense layer: 16 neurons, ReLU



Output layer: 1 neuron, sigmoid

MODEL COMPILATION

```
optimizer=tf.keras.optimizers.Adam(learning_rate=1e-3)  
loss="binary_crossentropy"
```

Adam is used as the optimizer. It adaptively changes the learning rate for each parameter and is commonly effective for neural-network training.

The loss is binary cross-entropy:

$$\text{BCE} = -[y \log(p) + (1-y) \log(1-p)]$$

where:

y = true label

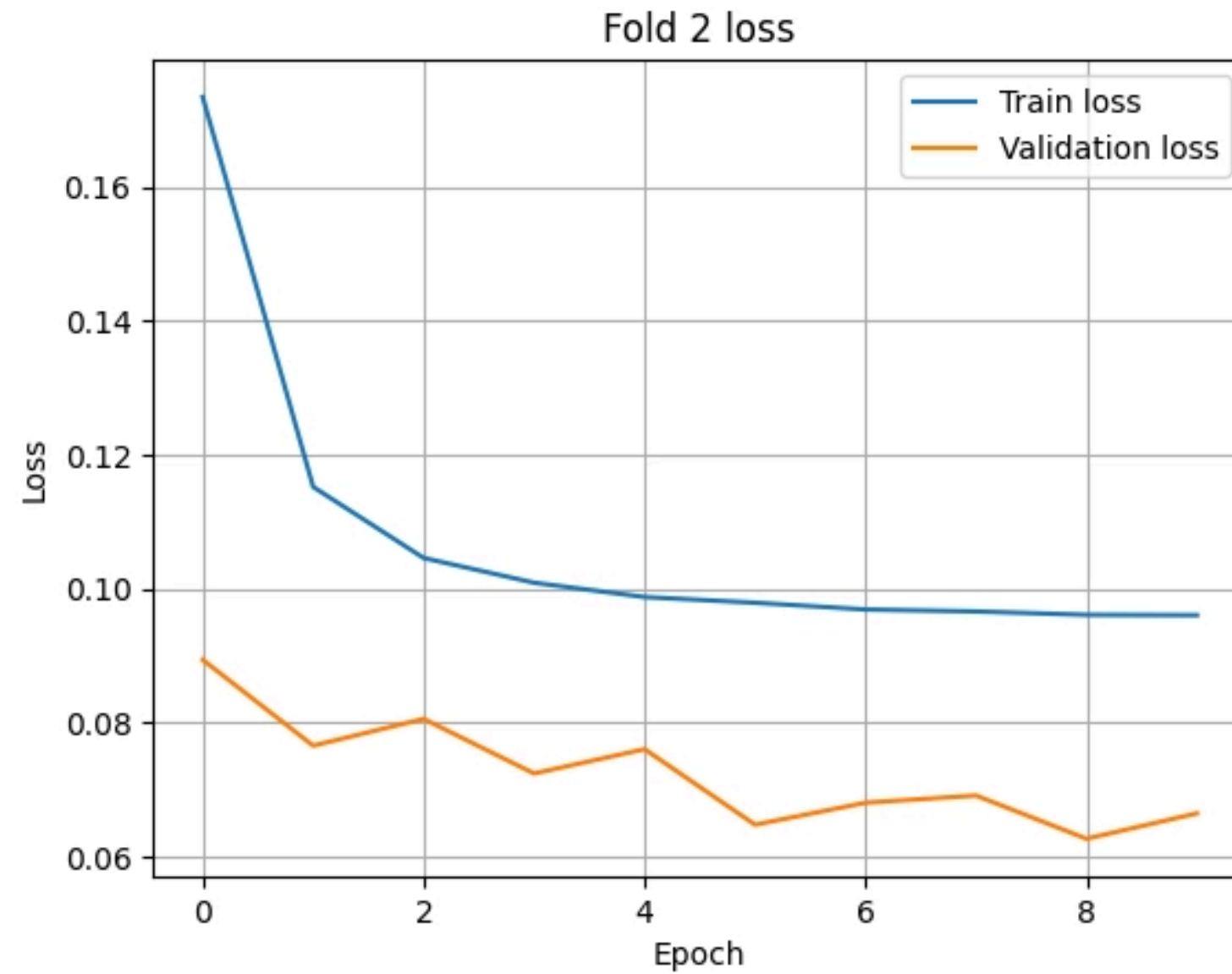
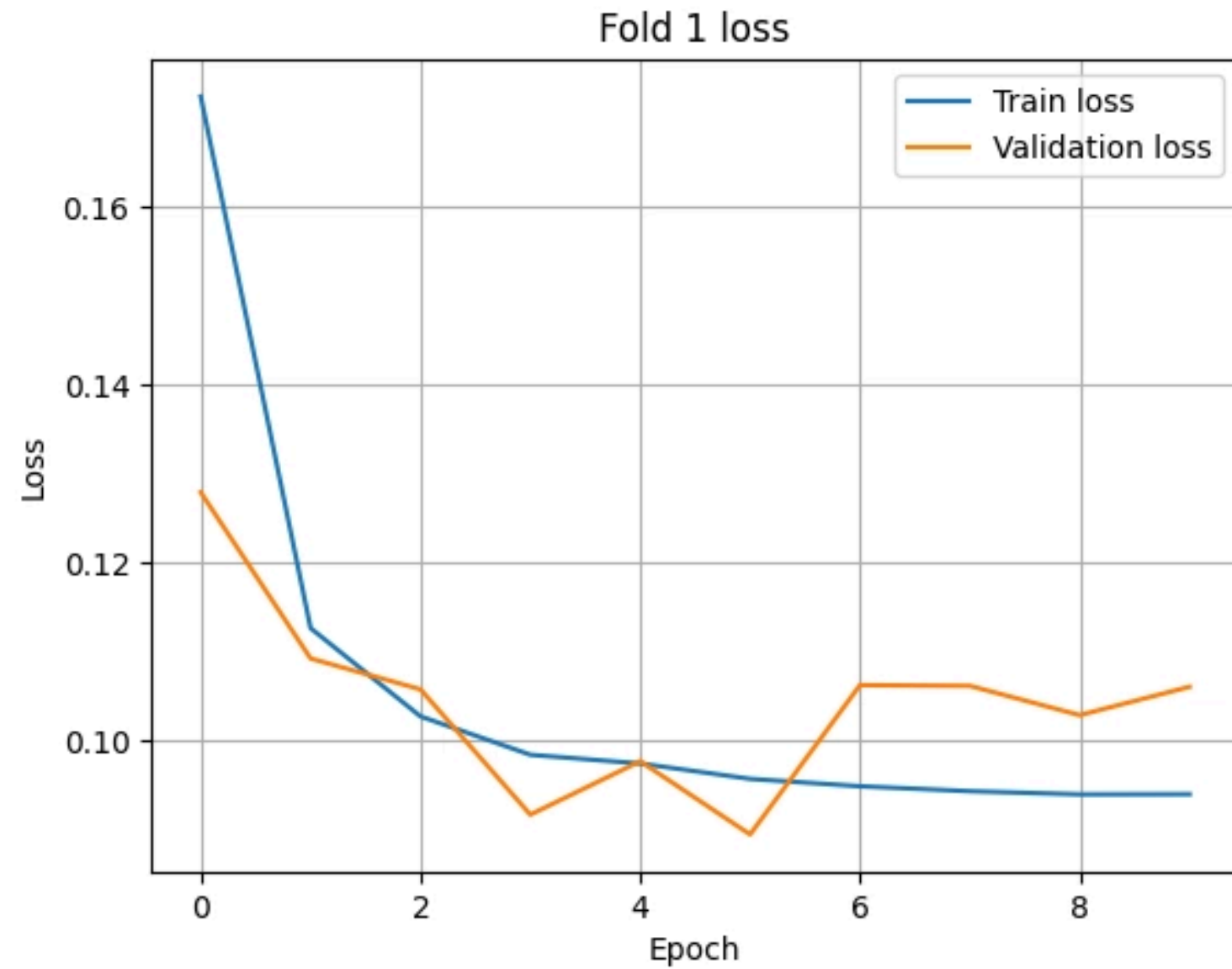
p = predicted WMH probability

METRICS

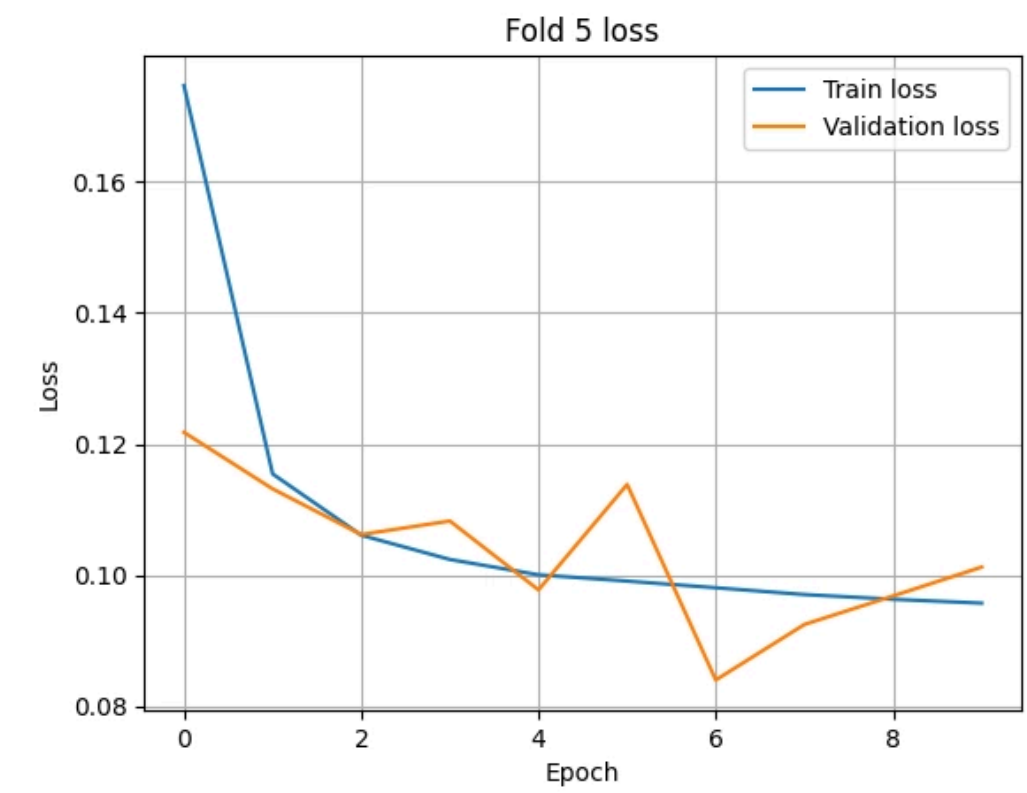
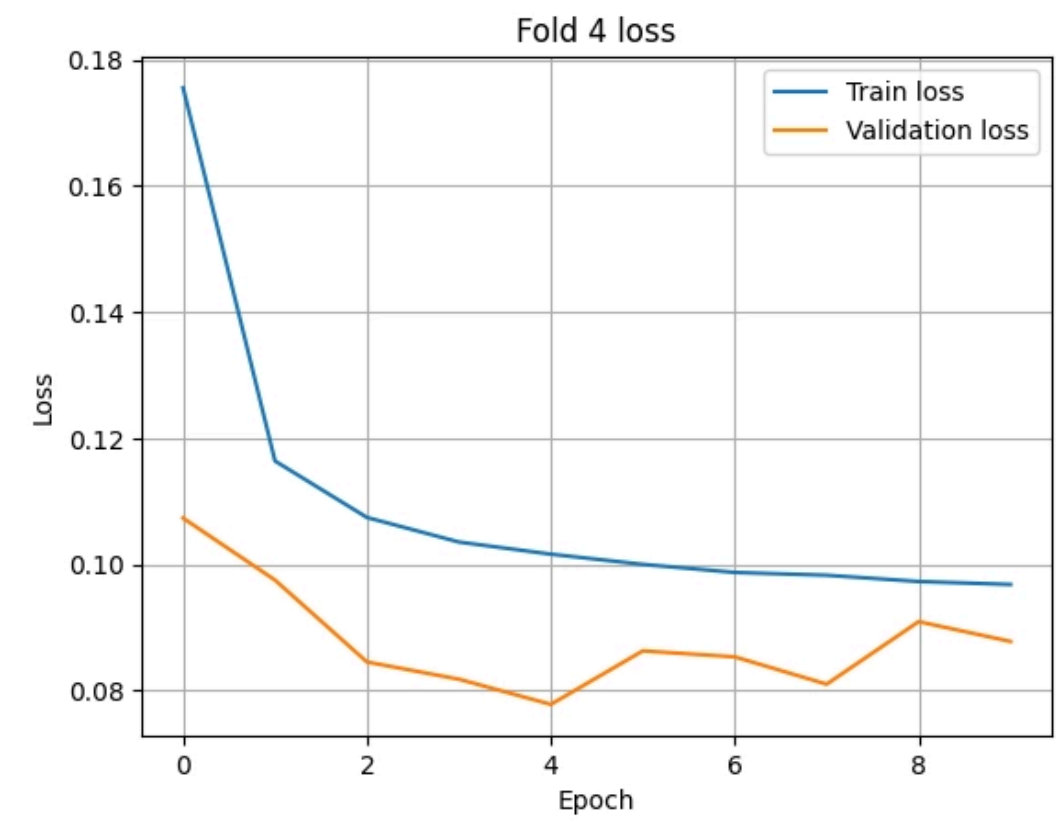
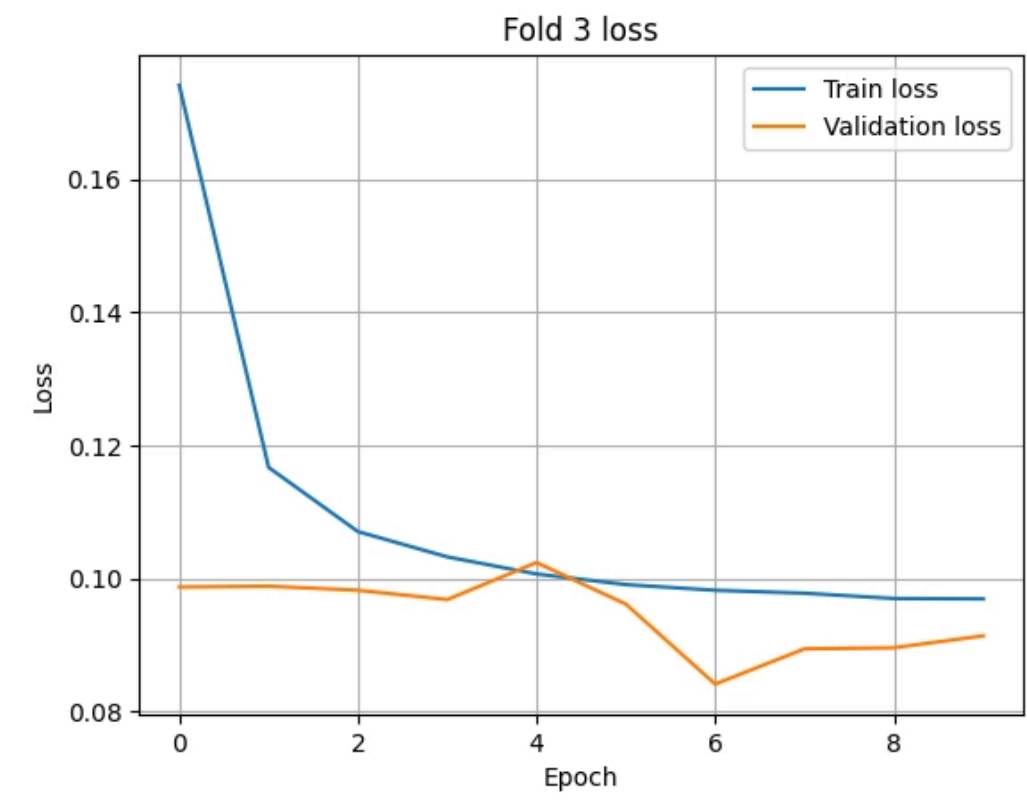
The metrics computed are the same as those computed in part a. The addition is the confusion matrix:

```
TEST RESULTS USING LOADED WEIGHTS
Dice:          0.4971
Sensitivity: 0.9497
Precision:    0.3367
Specificity: 0.9752
Confusion matrix:
[[2369761  60162]
 [   1616 30534]]
```

5-FOLD CROSS VALIDATION



5-FOLD CROSS VALIDATION



```
Load MATLAB preprocessed features
↓
Correct array dimensions
↓
Create one DataFrame row per brain pixel
↓
Keep patient_id to avoid leakage
↓
Split patients into train and test
↓
Analyze WMH/non-WMH imbalance
↓
Compute class weights
↓
Define a regularized feed-forward neural network
↓
Run GroupKFold cross-validation by patient
↓
Plot training/validation learning curves
↓
Train final model on all training patients
↓
Save model weights
↓
Reload weights
↓
Evaluate on independent test patients
```

PIPELINE OVERVIEW

**THANKS FOR THE
ATTENTION**